

Advanced (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $x(x + 3) + 2(x + 3)$
- (B) $x(x + 2) + 3(x + 2)$
- (C) $x(x + 3) + 2(x + 2)$
- (D) $(x + 2)(x + 3)$
- (E) $(x + 1)(x + 6)$

Q2. Factor the trinomial $x^2 - 7x + 12$.

- (A) $(x - 3)(x - 4)$
- (B) $(x - 2)(x - 5)$
- (C) $(x - 3)(x - 6)$
- (D) $(x - 1)(x - 6)$
- (E) $(x + 3)(x + 4)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 2)(x - 5)$
- (D) $(x - 2)(x + 5)$
- (E) $(x + 1)(x - 6)$

Q4. Factor the trinomial $x^2 - 4x - 21$.

- (A) $(x - 7)(x + 3)$
- (B) $(x + 7)(x - 3)$
- (C) $(x + 2)(x - 7)$
- (D) $(x - 2)(x + 7)$
- (E) $(x + 1)(x - 8)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $(x + 4)(x + 6)$
- (B) $(x + 2)(x + 8)$
- (C) $(x + 3)(x + 7)$
- (D) $(x + 1)(x + 9)$
- (E) $(x + 5)(x + 5)$

Q6. Factor the trinomial $x^2 - 2x - 35$.

- (A) $(x - 7)(x + 5)$
- (B) $(x + 7)(x - 5)$
- (C) $(x - 2)(x - 15)$
- (D) $(x + 2)(x + 15)$
- (E) $(x + 1)(x - 18)$

Q7. Factor the trinomial $x^2 + x - 12$.

- (A) $(x + 4)(x - 3)$
- (B) $(x - 4)(x + 3)$
- (C) $(x + 2)(x - 4)$
- (D) $(x - 2)(x + 4)$
- (E) $(x + 1)(x - 6)$

Q8. Factor the trinomial $x^2 - 9x + 20$.

- (A) $(x - 5)(x - 4)$
- (B) $(x + 5)(x + 4)$
- (C) $(x - 2)(x - 5)$
- (D) $(x + 2)(x + 5)$
- (E) $(x - 1)(x - 8)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $2x^2 + 5x - 3$.

Q10. Factor the trinomial $3x^2 - 2x - 1$.

Chef's Tips

- Check for common factors before factoring trinomials.
- Use the FOIL method for factoring trinomials of the form $(x + a)(x + b)$.
- Chef's Tips